

ACSAI
CALCULUS TUTORING

Worksheet 1: Prerequisites

Instructions: If you have any questions, need help, or you would like to propose an exercise that we could solve together in the next tutoring session, feel free to contact the tutor at *cela.2108230@studenti.uniroma1.it*.

Q. 1 Let $G = \{n \in \mathbb{Z} \mid n^2 < 10\}$, then

- a. $4 \in G$
- b. $-5 \in G$
- c. $G \cap \mathbb{N} = \{1, 2, 3\}$
- d. G has a finite number of elements (finite cardinality)

Q. 2 Determine which of the following numbers belongs to the set $\mathbb{Q} \setminus \mathbb{Z}$:

- a. 0
- b. $\frac{7}{3}$
- c. -5
- d. $\sqrt{4}$

Q. 3 Solve $|2x - 5| < 3$

- a. $x < 1$ or $x > 4$
- b. $1 < x < 4$
- c. $x < 4$
- d. $x > 1$

Q. 4 If $2^{2x-1} = 8$, find x

- a. $x = 1$
- b. $x = 2$
- c. $x = \frac{3}{2}$
- d. $x = \frac{5}{2}$

Q. 5 Let $E = \{\text{all prime numbers}\}$, then

- a. $1 \in E$
- b. $E \cap \mathbb{N} \neq \emptyset$
- c. $E \cap \mathbb{N} = \emptyset$
- d. E has a finite number of elements

Q. 6 The euclidean distance between points $A(1, 2)$ and $B(4, -2)$ is:

- a. 4
- b. 5
- c. $\sqrt{15}$
- d. 6

Q. 7 The inequality $-2 \leq x < 3$ corresponds to:

- a. $[-2, 3]$
- b. $(-2, 3]$
- c. $[-2, 3)$
- d. $(-2, 3)$

Q. 8 Quadrant with $x > 0, y < 0$:

- a. Quadrant I
- b. Quadrant II
- c. Quadrant III
- d. Quadrant IV

Q. 9 Consider the equation $x^2 - 2(\lambda + 1)x + \lambda^2 = 0$. For which values of $\lambda \in \mathbb{R}$ does the equation have exactly one real solution?

- a. $\lambda = 0$
- b. $\lambda = 1$
- c. $\lambda = -\frac{1}{2}$
- d. $\lambda = 2$

Q. 10 Solve $\frac{x+1}{2} \geq 3$:

- a. $x \geq 5$
- b. $x \leq 5$
- c. $x \geq 6$
- d. $x \leq 6$

Q.	A	B	C	D	Q.	A	B	C	D
1	☐	☐	☐	☐	6	☐	☐	☐	☐
2	☐	☐	☐	☐	7	☐	☐	☐	☐
3	☐	☐	☐	☐	8	☐	☐	☐	☐
4	☐	☐	☐	☐	9	☐	☐	☐	☐
5	☐	☐	☐	☐	10	☐	☐	☐	☐

Exercise 1. Solve $x^2(x^2 - 1) > 2x^4 + x^2$ and determine whether any solution intervals exist. Justify your answer.

Exercise 2. Solve the inequality and give the solution set:

$$\frac{x^2 - 5x + 6}{x^2 - 4} \leq 0$$

Exercise 3. Determine the solutions of the following inequality:

$$1 < ||x| - 1| < 2$$

Exercise 4. State whether the following sets are bounded, if so state whether they are bounded from above or below and determine (if they exist) the minimum and maximum of those sets.

a) $\{x \in \mathbb{R} \mid 4 < x^2 \leq 9\}$

b) $\{x \in \mathbb{R} \mid 4 \leq x^2 < 9\}$

c) $\{\frac{1}{n+3} : n \in \mathbb{N}\}$

d) $\{p^2 : p \in \mathbb{Z}\}$

Exercise 5. State whether the following statements are true or false.

1. $\forall q \in \mathbb{Q}, \exists p \in \mathbb{Q}$ such that $p^2 = q$ **T/F**

2. $\forall n \in \mathbb{N}, n^2 \geq n.$ **T/F**

3. $a \leq b, \iff a^3 \leq b^3$ **T/F**

4. $\forall x \in \mathbb{R}, x^2 > 0.$ **T/F**

5. $\exists p \in \mathbb{Q}, p^3 = 8.$ **T/F**

6. $\mathbb{N} \cap \mathbb{Z} = \mathbb{N}$ **T/F**

7. $\mathbb{R} \cap \mathbb{C} = \phi$ **T/F**

Exercise 6. (BONUS) Tangent Tom is a first-year ACSAI student. Unfortunately, he skipped his first calculus class and borrowed his friend's notes. Being a bit clumsy, he lost some of the sheets while trying to work through his homework. He stumbled upon a claim but the proof was in the sheets he lost.

The remaining notes show the following partial claim:

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Q} \text{ such that } x < y < x + 1 \text{ and } y \notin \mathbb{Z}.$$

Tangent Tom thought: "So every integer has a rational number strictly between it and the next integer that is not itself an integer... that seems obvious. Am I missing something?"

Question: Help Tangent Tom. Determine whether the claim is true or false. Justify your answer carefully. Explain why this statement might seem trivial at first glance, but actually tests your understanding of rational numbers, integers, and inequalities.

Hint: Let x be any integer and define $y = \frac{1}{2} + x$